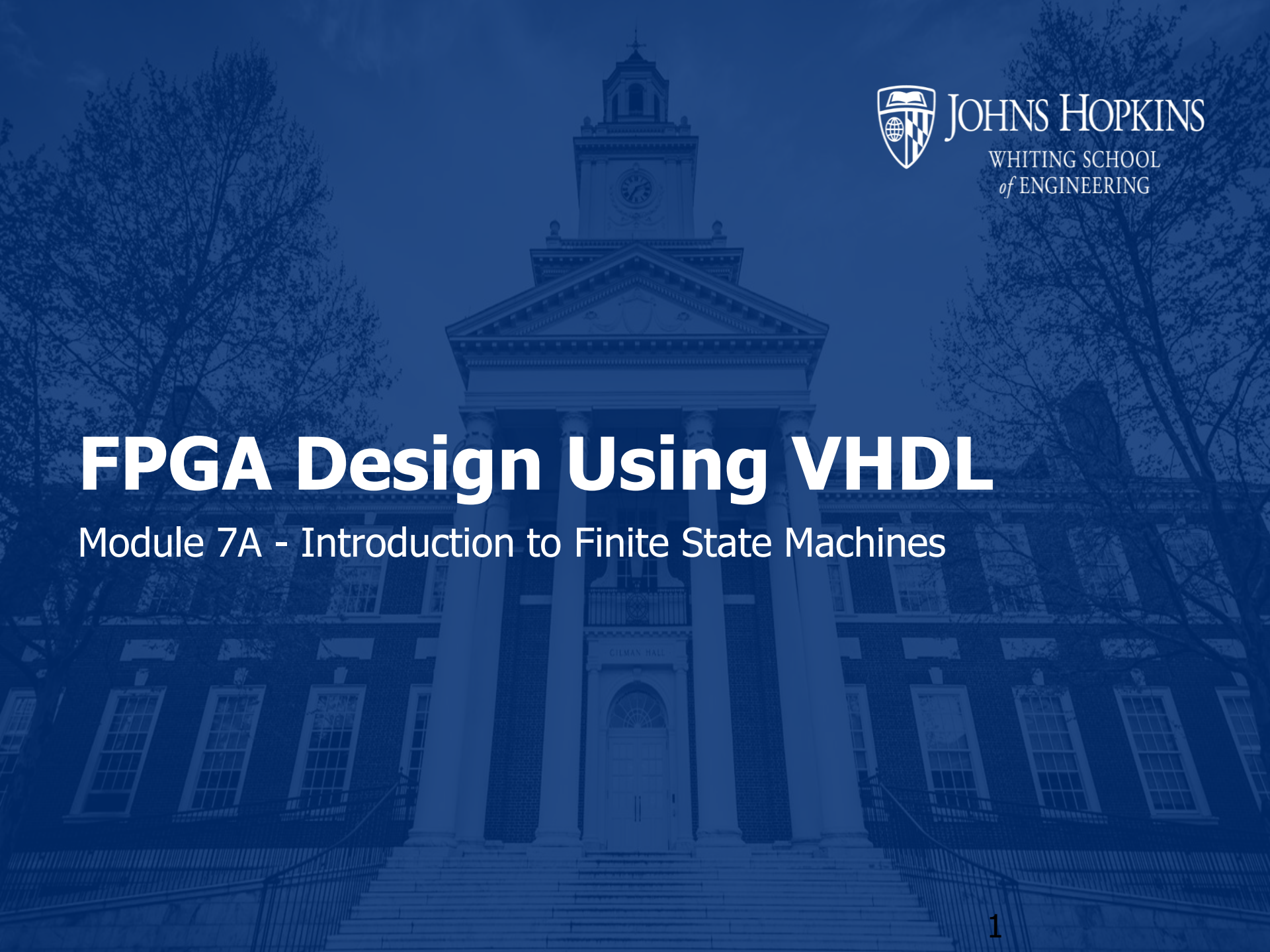


FPGA Design Using VHDL

Module 7A - Introduction to Finite State Machines

In This Module, You Should …

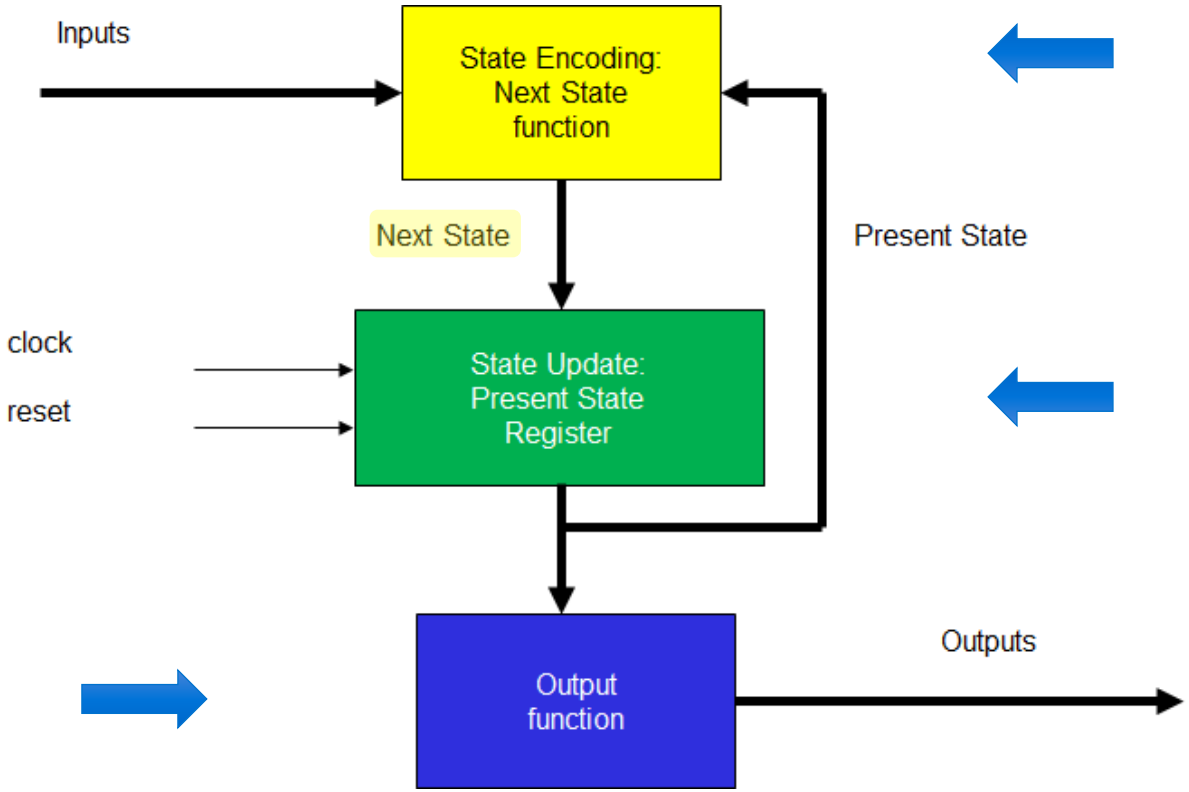
- Recall both types of Finite State Machines
- Draw state transition diagrams for both types of FSM
- Describe either type of FSM using **good synthesizable VHDL code**
- Construct a "pattern recognizer" using state transition diagram
- Explain how a FSM can "lock up" and how to design a "safe" FSM
- Construct an Serial Peripheral Interface (SPI) controller utilizing an FSM.

Finite State Machines (FSMs)

- Any circuit with memory is a FSM
  - Sequential counters are the simplest FSMs
- Design of FSMs involves
  - Defining inputs to the FSM
  - Defining states
  - Defining transitions between states
  - Defining the outputs
  - Optimization / minimization
- Two types of FSMs
  - Moore: Output depends on states only
  - Mealy: Output depends on states AND input(s)

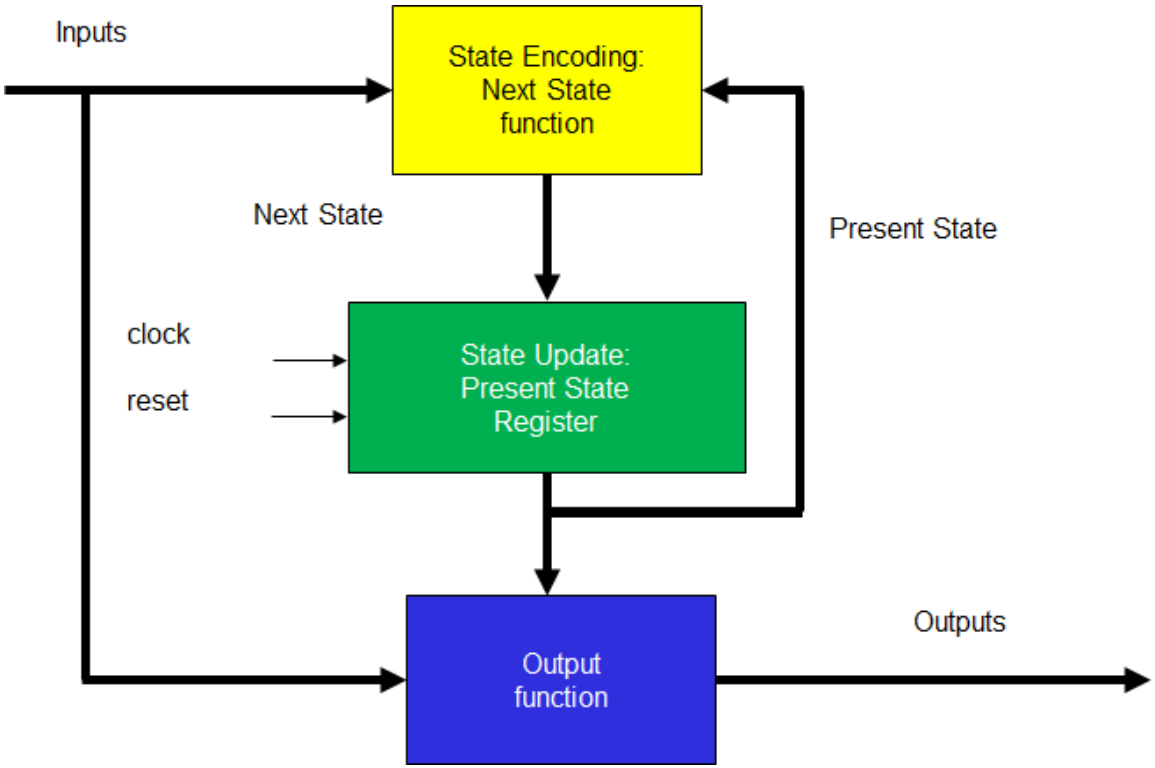
Moore FSM – Block Diagram

- Output is a function of present state only



Mealy FSM – Block Diagram

- Output is a function of a present state and inputs



Moore vs. Mealy FSM

- Moore and Mealy FSMs can be functionally equivalent
- Mealy FSM has richer description and usually requires smaller number of states
  - Moore FSM has "Moore" states than equivalent Mealy
- Mealy FSM computes outputs as soon as inputs change
  - Mealy FSM responds sooner than equivalent Moore FSM
- Moore FSM has no combinational path between inputs and outputs
  - Moore FSM is more likely to have a shorter critical path